

MICRODUCK

Pollen Robotics' latest robot release — a \$399 open-source biped you can train

DOSSIER ID: OSINT-2026-089 | SUBJECT: Microduck (Pollen Robotics SAS / Hugging Face) | STATUS: Pre-order — launched 27 Aug 2026

1 · EXECUTIVE SUMMARY

On **27 August 2026**, Pollen Robotics SAS — the Bordeaux (France) robotics subsidiary acquired by Hugging Face in April 2025 — opened pre-orders for **Microduck**, a 25 cm, ~780–800 g open-source biped robot at an introductory **\$399** (excl. taxes & shipping). It is Pollen Robotics' first mass-market product, following the \$70k Reachy 2 research humanoid (CES 2025) and the Reachy Mini desktop companion (2025, 10,000+ units shipped). Every behavior shipped in the box — walking, sitting, kicking, grasping, roller-skating, fall recovery — is a **neural policy trained in the MuJoCo physics simulator (mjlab + PPO) and deployed to the robot** through a fully open-source, Apache-2.0 stack (runtime, SDK, sim, RL recipes all on GitHub).

Demand was immediate and loud: Hugging Face co-founder **Thomas Wolf reported >\$1M in pre-orders within ~6 hours** of launch and **>\$2.6M within the first 24 hours**, creating a delivery backlog and reseller interest. Strategically, Microduck extends Hugging Face's "open-source physical AI" playbook (LeRobot, SO-101 arm, HopeJR) and lands while Hugging Face itself is reportedly in acquisition talks with Nvidia at a ~\$13B valuation — putting this small robot at the center of both the consumer-robotics and AI-industry spotlight.

PRODUCT	Microduck — open-source biped / RL learning platform
MANUFACTURER	Pollen Robotics SAS (Bordeaux, FR) — Hugging Face robotics team
LAUNCH	Announced 26–27 Aug 2026; pre-orders opened 27 Aug 2026
PRICE	\$399 introductory (excl. taxes/shipping); bulk pricing for 10+ units
SIZE / WEIGHT	25 cm tall · ~780–800 g
ACTUATION	15 motors (14 servo joints: legs/neck/head + articulated beak)
SENSORS	Wide-angle camera · 8×8 ToF LiDAR · 2× IMU · mics · speaker · 2× NFC (head & beak)
COMPUTE / POWER	Rockchip RK3566 + AI accel. · 1 GB RAM · 32 GB · NP-F550 battery (~1 h) · Wi-Fi/BT/USB-C
SOFTWARE	Apache-2.0: Rust runtime + SDK, mjlab RL stack; 50 Hz onboard policy loop
SKILLS IN BOX	7 trained behaviors (walk, sit/stand, kick, grab, roller-skate, stand-up, voice)
COLORWAYS	Cream · Graphite · Lavender · Sky
AVAILABILITY	US/CA/EU/UK/NO/CH/JP/KR at launch; deliveries targeted before Christmas 2026



Microduck (Cream colorway) — 25 cm biped with articulated grasping beak.
Source: pollen-robotics.com (Aug 2026)

■ **Bottom line:** Microduck is the cheapest production robot to date that ships with a complete, retrainable sim-to-real RL pipeline — a deliberate attempt to commoditize physical-AI experimentation and put “teachable” legged robots on every desk.

LAUNCH TIMELINE (ALL TIMES FROM OPEN SOURCES)

26 Aug 2026	Official launch blog post “Meet Microduck” published; microduck & microduck_rl repos released (Apache-2.0)
27 Aug 2026	Pre-orders open at \$399; product unveiled by Hugging Face CEO C. Delangue; >\$1M orders in ~6 h (T. Wolf, X)
28 Aug 2026	Sales reported >\$2.6M in first 24 h; coverage: TechCrunch, The Verge, Axios, The Register, Business Insider
By Christmas 2026	Target first deliveries — US/Canada/EU/UK/Norway/Switzerland/Japan/South Korea; more markets later

Naming note: company legal name is “Pollen Robotics SAS” (sometimes styled “Pollen Robotic”).

2 · TECHNICAL DEEP DIVE

2.1 Hardware platform

- **Compute:** Rockchip RK3566 SoC (quad Cortex-A55) with on-board AI accelerator; 1 GB RAM, 32 GB storage; Wi-Fi + Bluetooth; USB-C.
- **Actuation:** 15 motors — 14 servo joints (10 leg: hip_yaw/roll/pitch, knee, ankle per side; 4 neck/head: neck_pitch, head_pitch/yaw/roll) plus an articulated grasping beak; closed-loop ~50 Hz control.
- **Perception:** wide-angle front camera; compact 8×8 ToF LiDAR; 2 IMUs; microphones + speaker; 2 NFC antennas (head and beak) for tag-triggered behaviors.
- **Power & mechanics:** removable NP-F550 battery (~1 h runtime); ~800 g total; four snap-on shell colorways; self-recovery from falls; passive-wheel roller-skate accessory.

2.2 Onboard software & open-source architecture

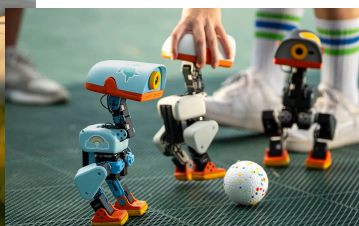
- **Apache-2.0** across the stack: *pollen-robotics/microduck* (runtime + SDK) and *pollen-robotics/microduck_rl* (training); 3D models CC BY-SA-NC; 214 public repos org-wide.
- **Rust daemon design** over one JSON-RPC contract on Unix sockets: **robotd** (50 Hz control loop + motor bus), **updaterd** (signed, health-gated, reversible updates), **configd** (Wi-Fi/identity), **btd** (Bluetooth), **padd** (gamepad), **mediad** (WebRTC camera stream), **tofd** (depth sensor).
- **CLI & tooling:** *robotctl* (status/configure/update), *duckctl* (drive from laptop over Bluetooth), in-repo cheat sheets and design docs.
- **Privacy stance:** local-first, open camera/mic stack; CEO Delangue contrasts open-source robots with “black-box” cloud systems — though third-party apps can still access sensors.

2.3 Reinforcement learning & sim-to-real stack

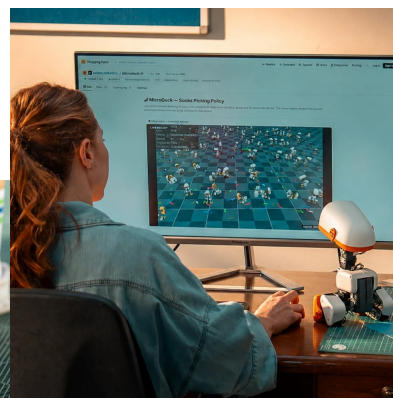
- **Training:** mjlab framework (MuJoCo + Warp) with PPO; policies trained at 50 Hz and exported to **ONNX** (observation normalizer baked in) for the onboard runtime.
- **Sim2real recipe:** BAM actuator model for the Dynamixel XL330 (voltage control law, back-EMF, Coulomb/Stribeck friction), per-env domain randomization (battery voltage, voltage sag, command delay, friction), and $\pm 1^\circ$ gear-backlash modeling with output-side encoder feedback.
- **Shared 61-dim observation contract** (48 proprioception + twist/head/body commands) lets the runtime **hot-swap walk, recover and trick policies** at any moment.
- **Compute footprint:** usable walking gait in ~1–2 h at 4,096 parallel envs on CUDA; cloud training via Hugging Face Jobs (`--hf-jobs`); 13+ published task environments incl. flat/rough and roller variants.
- **Sharing model:** the 7 shipping policies are published and retrainable; behaviors are meant to circulate on the HF Hub “models are useful because people build on one another’s work.”



Roller-skate mode



Kick / football behavior



Simulation twin (MuJoCo)

SHIPPED BEHAVIORS — 7 TRAINED POLICIES (ALL RETRAINABLE)

Behavior	What it does	Published policy (microduck_rl)
Walk	Velocity-tracking gait, gamepad-driven; head-pose commands	Mjlab-Velocity-Flat/Rough
Sit & stand	Commanded sit↔stand in one policy, gentle, head commandable	Mjlab-SitStand
Kick	One-shot kick of 70 mm / 15 g ball, then back to walking	Mjlab-BallKick
Grab & carry	Whole body lowers, beak scoops, pops back upright	Mjlab-GroundPick
Roller skating	Locomotion on passive-wheel skates (crouch, spin, slope, swizzle variants)	Mjlab-*-Rollers
Get back up	Stands from face-down / face-up / sitting; self-recovery	Mjlab-StandUp / VelStand
Voice & tricks	Unique per-unit generated voice; laser-follow; NFC-triggered moves	Runtime (onboard)

Joint layout (14 servos): 0–4 left leg (hip_yaw, hip_roll, hip_pitch, knee, ankle), 5–8 neck/head, 9–13 right leg; 15th motor drives the beak. 3D models exported from Onshape; backlash variants of every task exist for robustness.

3 · COMMERCIAL & MARKET INTELLIGENCE

3.1 Launch performance & demand signals

- **>\$1M in ~6 hours** (27 Aug) and **>\$2.6M in first 24 hours** (28 Aug) of pre-orders — reported by Hugging Face co-founder Thomas Wolf on X; backlog formed almost immediately.
- **Reseller & institutional pull:** distributors ordering units; bulk pricing, dedicated support and import documentation offered for 10+ units (education/research).
- **Positioning:** “Welcome to the era of open-source affordable robots to democratize physical AI and world models!” — Hugging Face CEO Clem Delangue.

3.2 Pricing & SKUs (USD, excl. taxes/shipping)

SKU	Price	Contents
Microduck (robot)	\$399	Robot, battery, USB-C cable, game controller — 4 colorways
Charger pack	\$39	Dual charger, 2× batteries
Dev pack	\$119	3× spare motors, 5× motor cables, 2× batteries, dual charger, 10× NFC tags, Hugging Face credit, screwdriver + screws
Accessory pack	\$39	Laser pointer, NFC polaroid, 2× rollers, ball, 10× NFC tags
Context — Reachy Mini	\$499 / \$399 Lite	Pollen’s desktop companion robot (2025); Lite version runs off a host Mac/PC

3.3 Strategic context: company & ecosystem

- **Company:** founded May 2016 in Bordeaux by Matthieu Lapeyre & Pierre Rouanet (ex-Inria “Flowers” team; creators of the open-source Poppy robot). ~\$2.6M raised (EU / EIC Fund, Hugging Face); acquired by Hugging Face in April 2025 (undisclosed).
- **Product lineage:** Reachy (2019–2020) → Reachy 2 (\$70K open-source humanoid, CES 2025; used by Cornell & Carnegie Mellon labs) → Reachy Mini (2025, 10,000+ shipped) → Microduck (Aug 2026).
- **Ecosystem:** Hugging Face LeRobot (2024), SO-101 arm and HopeJR humanoid (May 2025), HF Hub/Spaces/Jobs; Microduck treats robot policies like model weights — train in sim, publish, redeploy; active Discord community (“Join the flock”).

3.4 Competitive landscape & risk watch

- **Price disruption:** \$399 vs \$70K Reachy 2 collapses the cost floor for legged-RL research; undercuts most hobby/lab bipeds and complements (rather than competes with) HF’s Reachy line.
- **Incumbents:** enterprise humanoid makers (Unitree, Agility Robotics, Appteronik — 100+ tracked competitors) target industry; Microduck owns education/maker/consumer tier — with no direct open-source rival at this price.
- **Risks — privacy:** camera + mics inside homes; open source mitigates opacity but third-party apps can still access sensors; **fulfillment:** Christmas 2026 promise vs 24-h backlog and small team; **parent volatility:** reported Nvidia–Hugging Face acquisition talks (~\$13B) and a recent HF security incident (OpenAI sandbox breach during safety testing); **technical limits:** armless, ~1 h battery, small payload — sim2real quality depends on user skill.

3.5 Intelligence assessment (key judgments)

- **J1 — (High confidence)** Microduck is positioned to become the default low-cost entry point for sim-to-real and legged-RL education — the “Raspberry Pi of physical AI.”
- **J2 — (High confidence)** The \$2.6M/24-h figure (~ 5,000–6,500 units incl. packs) is the strongest consumer-demand signal yet recorded for open-source robot hardware; expect production to sell out repeatedly.
- **J3 — (Medium-high)** Community policy proliferation on the HF Hub will create the first practical “app store for robot behaviors” — and with it, new safety/liability debates over user-uploaded policies running in homes.
- **J4 — (Medium)** If the Nvidia–Hugging Face deal closes, expect Microduck to be folded into Nvidia’s robotics stack (Jetson/Isaac), sharply expanding its reach and hardening its supply chain.
- **J5 — (Medium)** Nearest-term risk is fulfillment: watch for shipping-date slips or early QC issues; hardware + RL tooling quality in first deliveries will set long-term reputation.

3.6 Sources & methodology

Primary: pollen-robotics.com (product page, launch blog “Meet Microduck”, 26–27 Aug 2026); store.pollen-robotics.com (specs/pricing); GitHub org pollen-robotics (microduck, microduck_rl, mjlal); official X posts (C. Delangue, T. Wolf). **Secondary:** TechCrunch (27 Aug), The Verge (27 Aug), Axios (27 Aug), The Register (27 Aug), Business Insider (28 Aug), MarkTechPost, Yahoo/Mashable, RoboAtlas, RoboBoom, Origin of Bots, Tracxn, The Company Check. **Method:** open-source collection, cross-verification and product/repo inspection on 29 Aug 2026. All prices USD before taxes/shipping; figures as reported by third parties. **Disclaimer:** compiled exclusively from open sources for informational purposes; not affiliated with or endorsed by Pollen Robotics or Hugging Face.



The four colorways: Cream · Graphite · Lavender · Sky. Source: store.pollen-robotics.com